

Maggie Rouin

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Education

Professional Science Masters, Ecosystem Science and Sustainability

Colorado State University – Fort Collins, CO

Dec. 2026

Bachelor of Science, Environmental Resource Management

The Pennsylvania State University – University Park, PA

May 2021

Industry Experience

SWCA Environmental Consultants – Assistant Project Environmental Scientist

Jun. 2022 – May 2025

- Led and assisted with project management tasks, such as writing proposals, budgeting, client coordination, and managing technical tasks for various projects, including water resources and renewables projects.
- Wrote environmental assessment reports which detail the findings of field surveys and convey pertinent environmental regulations to our clients.
- Analyzed and QA/QC field data from site visits to interpret in environmental assessment reports.
- Conducted site assessment surveys to identify sensitive environmental resources on project sites.

CMS Environmental Solutions – Environmental Analyst

Nov. 2021 - Jun. 2022

- Inspected 40-50 commercial, residential, and industrial construction sites for environmental compliance.
- Wrote inspection reports detailing site findings which were then used to correct pollutant discharges.
- Routinely communicated with clients and regulators to relay inspection findings and discuss solutions for pollution prevention.
- Reviewed and audited client stormwater pollution prevention plans to ensure that their plans were compliant with COR400000 and Denver city regulations.

Penn State Sustainable Ag, Food, and Environmental Science (SAFES) Institute – Environmental Intern

Jun.– Sept. 2021

- Analyzed GIS mapping data and conducted site visits, with the County Conservation District, of a local watershed for the Chesapeake Bay Foundation through ArcGIS to identify current and future sites for BMP implementation.
- Responsible for creating, leading, and obtaining partners for two stakeholder education workshops, one for the Homeowner's Guide to Stormwater Management and another for a stream health citizen science tool, FISH.
- Conducted water and fish samplings in local watersheds to analyze water quality and stream health following best management practice (BMP) implementation.

Penn State Ag and Environment Center – Water Quality Intern

Jan.– May 2021

- Conducted literary reviews on the effectiveness of stream, wetland, and floodplain BMPs, compiled summaries, and presented results.
- Analyzed GIS mapping data of three local watersheds through Google Earth Pro and ArcGIS to identify current and future sites for BMP implementation.
- Conducted site visits, based on GIS assessments, to determine the viability of potential BMP implementation locations.

U.S. Environmental Protection Agency – Office of Compliance Intern

Jun.- Aug. 2019

- Tracked, analyzed, and presented trends in data from regional EPA National Pollutant Discharge Elimination System (NPDES) Compliance Monitoring Strategy (CMS) reports.
- Collaborated with two team members to learn wiki markup code to edit the EPA's Stormwater Inspector Training Library webpage for stakeholder use.

- Redesigned the EPA's Drinking Water Compliance Resource Center SharePoint site for consistency.

Nuclear Energy Institute (NEI) - Governmental Affairs Intern

Jun. - Aug. 2018

- Attended Congressional committee hearings pertinent to NEI and composed hearing reports.
- Organized the company's Political Action Committee (PAC) database.
- Collaborated with the Governmental Affairs division to complete routine office tasks.

Academic Experience

Penn State Department of Plant Science- Research Assistant

Sept. 2018 - Jan. 2019

- Scanned 150-200 Phaseolus roots to support the research of stress tolerance mechanisms in plant roots.
- Organized and labeled the scanned roots according to phenotype and phosphorous levels.

Penn State Department of Biology- Assistant Lab Teaching Assistant

Aug. - Dec. 2019

- Assisted 40 students and 2 TAs with lab set up, procedures, and instruction for BIO 220W Lab.
- Attended weekly meetings to discuss upcoming lab content and assignments.

Coursework

Integrated Ecosystem/Sustainability Analysis - ESS 500

Fall 2025

- Introduction to concepts and techniques that underpin future learning related to water resources, carbon management, sustainable food systems, environmental data science, and climate resilience.

Environmental Data Science Application – ESS 523a

Fall 2025

- Explored tools and best practices for working with large environmental datasets using the programming language R.
- Utilized iteration, functional programming, debugging, code management, and collaboration with git and GitHub.

Environmental Data Science Application: Water Resources – ESS 523c

Spring 2026

- Analyzing and understanding water resources through key innovations in deep learning for hydrological prediction and model parameterization, with a focus on cutting-edge techniques and hands-on analyses.

Environmental Data Science Applications: Remote Sensing – ESS 523d

Spring 2026

- Practical methods related to open science remote-sensing analysis, with a focus on coding-derived remote sensing data processing and machine learning analysis in a reproducible research framework.
- Utilizing R for statistical analysis and open science processes.

Land Use Hydrology – WR 416

Fall 2025

- Fundamental concepts in hydrology and effects of land use on hydrologic processes.

Land Use and Water Quality – WR 418

Spring 2026

- Physical, chemical, biological water quality parameters affecting land use; land management to maintain water quality; water quality standards, legislation.

Water Law – WR 512

Fall 2025

- Basics of water law and policy for Colorado, western states, and the U.S.

Watershed Problem Analysis – WR 696

Spring 2026

- Research project development for spatial watershed issues, focused on problem solving in watershed science.